MATH6006: Statistical Methods for OR Modelling

Christine Currie

Mathematical Sciences

University of Southampton

2026/2027

Discrete random variables and distributions

We consider the next level of statistical analysis: fitting data to a *parametric distribution*.

Discrete random variables and distributions

We consider the next level of statistical analysis: fitting data to a *parametric distribution*. In this section we cover:

- Random variables and probability
- Conditional probabilities
- Popular discrete probability distributions
- Moment generating functions

Random variables and Probability

A process is *random* if individual outcomes are uncertain.
The *probability* of any outcome of a random process is the proportion of times that outcome would occur as the number of repetitions tends to infinity.

Tossing a Coin

- Toss a coin once, there are two possible outcomes: H or T.
- Toss a coin twice, there are four possible outcomes: HH, HT, TH, TT.
- Toss a coin three times, there are eight possible outcomes: HHH, HHT, HTH, THH, THT, TTH, HTT, TTT.

Comments on Coins

(i) If we toss a coin, the outcome is unpredictable.

(ii) If we toss a coin many times, the number of Heads $\approx$ the number of Tails. The *probability* of an H is about $1/2$.

(iii) The sample space S is a set of all the possible events, but also any subset of S is an event.

The famous statistician Karl Pearson once tossed a coin 24000 times, recording 12012 times of H appearing, a proportion of 0.5005, which is very close to $1/2$.

Discrete Probability Model

Definition

A *discrete random variable* X with values $x_0, x_1, x_2, \ldots$ is described by its probability distribution function $p(k)$,

$$p(k) = P\{X = x_k\}, \qquad k = 0, 1, 2, 3, \ldots$$

where $0 \leq p(k) \leq 1$, $\sum_{k=0}^{\infty} p(k) = 1$.

Discrete Probability Model

Definition

A *discrete random variable* X with values $x_0, x_1, x_2, \ldots$ is described by its probability distribution function $p(k)$,

$$p(k) = P\{X = x_k\}, \qquad k = 0, 1, 2, 3, \ldots$$

where $0 \leq p(k) \leq 1$, $\sum_{k=0}^{\infty} p(k) = 1$.

The sample space $S = \{x_1,\ x_2,\ x_3, \ldots\}$.
The probabilities are $P = \{p_1,\ p_2,\ p_3, \ldots\}$, where $p_k = P(X = x_k)$.

Notation Reminder

- Capital leter, e.g. X, denotes the random variable.
- Lowercase letter, e.g. x, denotes a possible value.

Cumulative Distribution Function for a Discrete Random Variable

For a discrete random variable, the *cumulative distribution function* $F(x)$ of X is defined as

$$F(x) = P\{X \leq x\} = \sum_{\{k|x_k \leq x\}} P\{X = x_k\} = \sum_{\{k|x_k \leq x\}} p(k).$$

Conditional Probability

If A and B are two events, the probability that B happens given that A has happened is denoted as

$$P(B|A) = \frac{P(A \cap B)}{P(A)}.$$

Conditional Probability

If A and B are two events, the probability that B happens given that A has happened is denoted as

$$P(B|A) = \frac{P(A \cap B)}{P(A)}.$$

If the occurrence or non-occurrence of A does not affect the probability of occurrence of B, then

$$P(B|A) = P(B),$$

and we say that A and B are *independent* events. Otherwise they are *dependent*.

Conditional Probability

If A and B are two events, the probability that B happens given that A has happened is denoted as

$$P(B|A) = \frac{P(A \cap B)}{P(A)}.$$

If the occurrence or non-occurrence of A does not affect the probability of occurrence of B, then

$$P(B|A) = P(B),$$

and we say that A and B are *independent* events. Otherwise they are *dependent*.

If A and B are independent, then

$$P(A \cap B) = P(A) \times P(B).$$

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.
The probability that A happens is

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.

The probability that A happens is

$$P(A) = \frac{4}{4+2} = \frac{2}{3}.$$

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.
The probability that A happens is

$$P(A) = \frac{4}{4+2} = \frac{2}{3}.$$

The probability that B happens given that A has happened is

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.
The probability that A happens is

$$P(A) = \frac{4}{4+2} = \frac{2}{3}.$$

The probability that B happens given that A has happened is

$$P(B|A) = \frac{3}{3+2} = \frac{3}{5}.$$

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.
The probability that A happens is

$$P(A) = \frac{4}{4+2} = \frac{2}{3}.$$

The probability that B happens given that A has happened is

$$P(B|A) = \frac{3}{3+2} = \frac{3}{5}.$$

The probability that both balls drawn are white is

Example: Independence

Suppose that a bag contains 4 white balls and 2 black balls. Let A be the event *first ball drawn is white* and B be the event *second ball drawn is white*, where the balls are not replaced after being drawn. Here, A and B are dependent events.
The probability that A happens is

$$P(A) = \frac{4}{4+2} = \frac{2}{3}.$$

The probability that B happens given that A has happened is

$$P(B|A) = \frac{3}{3+2} = \frac{3}{5}.$$

The probability that both balls drawn are white is

$$P(A \cap B) = P(A) \times P(B|A) = \frac{2}{3} \times \frac{3}{5} = \frac{2}{5}.$$

Expectation

The *expectation* or the *expected value* of a discrete random variable X with possible values $x_0, x_1, x_2, \ldots$ is defined by

$$\mu_X = \mathbf{E}[X] = \sum_{k=0}^{\infty} x_k P\{X = x_k\}.$$

The expectation is also called the *mean* of a random variable X and is denoted by $E[X]$, μ_x or μ.

Expectation

The *expectation* or the *expected value* of a discrete random variable X with possible values $x_0, x_1, x_2, \ldots$ is defined by

$$\mu_X = \mathbf{E}[X] = \sum_{k=0}^{\infty} x_k P\{X = x_k\}.$$

The expectation is also called the *mean* of a random variable X and is denoted by $E[X]$, μ_x or μ.

Note that the sum $\sum_{k=0}^{\infty} x_k P\{X = x_k\}$ represents the weighted average of X where each value x_k of X is weighted by $p(k) = P\{X = x_k\}$.

Variance

The variance of a random variable X with possible values $x_0, x_1, x_2, \ldots$ is defined by

$$\sigma_X^2 = Var(X) = \mathbf{E}(X - \mathbf{E}[X])^2.$$

By expanding the right hand side we get

$$Var(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2.$$

an expression that is more convenient for calculations.

Functions of a Random Variable

Assuming that X is a discrete random variable and $g(\cdot)$ a function, $g(X)$ is a random variable and

$$\mathbf{E}[g(X)] = \sum_{k=0}^{\infty} g(x_k) P\{X = x_k\}.$$

For example, where $g(x) = (x - E[X])^2$, we have:

$$\begin{aligned} E(g(X)) &= E[(X - E[X])^2] && (1)\\ &= \sum_{k=0}^{\infty} (x_k - E[X])^2 P\{X = x_k\} && (2)\\ &= \sum_{k=0}^{\infty} x_k^2 P\{X = x_k\} - (E[X])^2 && (3)\\ &= E[X^2] - (E[X])^2. && (4) \end{aligned}$$

Properties of the Variance

Hence,

$$Var(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

The variance of X, $Var(X)$ is often denoted by σ^2 or σ_X^2.

The standard deviation σ of X is defined as the square root of the variance:

$$\sigma = \sigma_X = \sqrt{Var(X)}$$

Properties of the Variance

Hence,

$$Var(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

The variance of X, $Var(X)$ is often denoted by σ^2 or σ_X^2.

The standard deviation σ of X is defined as the square root of the variance:

$$\sigma = \sigma_X = \sqrt{Var(X)}$$

It is easy to verify the following properties of the variance:

- $Var(aX + b) = a^2 Var(X)$, where a, b are constants;
- $Var(X + Y) = Var(X) + Var(Y)$ when X and Y are independent random variables.

Parametric Discrete Distributions

The most commonly used discrete distributions are:

- Discrete uniform distribution
- Bernoulli distribution
- Binomial distribution
- Negative Binomial distribution
- Geometric distribution
- Hypergeometric distribution
- Poisson distribution

For each of the listed distributions we will write down its probability distribution function, give a basic example and calculate the expectation and the variance.

Discrete Uniform Distribution

Definition

A discrete random variable X has a **uniform** distribution if it assumes each of its finitely many values $x_1, x_2, ..., x_n$ with the same probability,

$$P\{X = x_k\} = \frac{1}{n}, \qquad k = 1, 2, ..., n.$$

Discrete Uniform Distribution

Definition

A discrete random variable X has a **uniform** distribution if it assumes each of its finitely many values $x_1, x_2, ..., x_n$ with the same probability,

$$P\{X = x_k\} = \frac{1}{n}, \qquad k = 1, 2, ..., n.$$

The expected value is the average of the possible values:

$$\mu_X = \mathbf{E}(X) = \sum_{k=1}^{n} x_k P\{X = x_k\} = \frac{1}{n} \sum_{k=1}^{n} x_k = \bar{x}$$

and the variance is

$$\sigma_X^2 = E[(X - \mu_X)^2] = \sum_{k=1}^{n} x_k^2 p_k - \mu_X^2 = \frac{1}{n} \sum_{k=1}^{n} x_k^2 - \bar{x}^2.$$

Example: Tossing a Die

$$P\{X = x_k\} = \frac{1}{6}, \qquad k = 1, 2, ..., 6.$$

Its expectation is

Example: Tossing a Die

$$P\{X = x_k\} = \frac{1}{6}, \qquad k = 1, 2, ..., 6.$$

Its expectation is

$$\mu_X = \frac{1}{n}\sum_{k=1}^{6} k = \frac{1+2+3+4+5+6}{6} = 3.5.$$

Note that the expectation is NOT one of the outcomes in x_k.

Bernoulli random variable

Definition

A **Bernoulli** random variable X is a random variable with two outcomes, $X \in \{0, 1\}$, and

$$P\{X = 1\} = p, \quad P\{X = 0\} = q = 1 - p.$$

Bernoulli random variable

Definition

A **Bernoulli** random variable X is a random variable with two outcomes, $X \in \{0, 1\}$, and

$$P\{X = 1\} = p, \quad P\{X = 0\} = q = 1 - p.$$

The mean is

$$\mu_X = E(X) = \sum_x xP(x) = 0 \times (1 - p) + 1 \times p = p$$

and the variance is

$$\begin{aligned}\sigma_X^2 &= E[(X - \mu_X)^2] \\ &= \sum_x x^2 P(x) - \mu_X^2 = 0^2 \times (1 - p) + 1^2 \times p - p^2 \\ &= p(1 - p) = pq.\end{aligned}$$

e.g. tossing a coin.

Binomial random variable

Definition

A random variable X is a **binomial** random variable with parameters n and p, $0 < p < 1$, if

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, 2, ..., n$$

Binomial random variable

Definition

A random variable X is a **binomial** random variable with parameters n and p, $0 < p < 1$, if

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, 2, ..., n$$

If X is a binomial random variable, we write $X \sim$ bin(n,p).

More on the Binomial Distribution

Recall the binomial theorem:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} \qquad \text{with} \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Let $a = p$ and $b = 1 - p$, then the binomial theorem implies

$$\sum_{k=0}^{n} P\{X = k\} = (1 - p + p)^n = 1^n = 1.$$

Example: Tossing a Coin

Let X be the number of heads in n tosses of a fair coin, where the probability of a head in each single toss is $p = 1/2$. Possible values for X are $0, 1, 2, 3, ..., n$ and the probability of k heads is given by

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k} = \binom{n}{k} \frac{1}{2^n}, \qquad k = 0, 1, 2, ..., n$$

where $\binom{n}{k}$ is the number of different arrangements of k heads in n tosses.

Example: Tossing a Coin

Let X be the number of heads in n tosses of a fair coin, where the probability of a head in each single toss is $p = 1/2$. Possible values for X are $0, 1, 2, 3, ..., n$ and the probability of k heads is given by

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k} = \binom{n}{k} \frac{1}{2^n}, \qquad k = 0, 1, 2, ..., n$$

where $\binom{n}{k}$ is the number of different arrangements of k heads in n tosses.

The probability that an outcome will occur k times in n performances follows a binomial distribution with parameters n and p. A binomial variable is thus the result of repeating the Bernoulli model n times.

Expectation and Variance of a Binomial Random Variable

The expectation and the variance of a binomial random variable X can be calculated as follows. We can think of X as the sum of n independent Bernoulli variables, $X_1, X_2, ..., X_n$,

$$X = \sum_{i=1}^{n} X_i$$

and using this we get

$$\mathbf{E}(X) = \mathbf{E}\left(\sum_{i=1}^{n} Xi\right) = \sum_{i=1}^{n} \mathbf{E}(X_i) = np.$$

Similarly, for the variance we have:

$$Var(X) = Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) = npq.$$

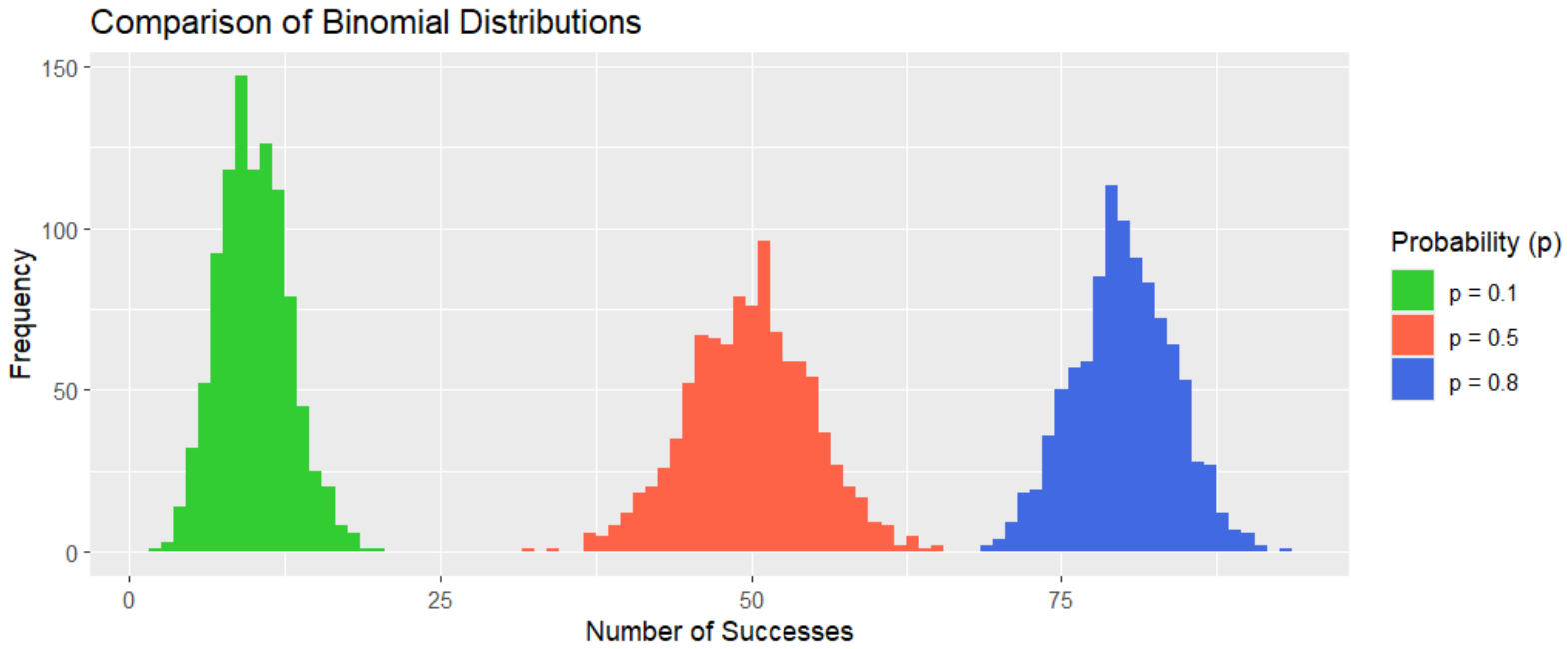
Note that for $Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i)$ we need to assume that the X_i are independent.

Simulating and Plotting a Binomial

```r
# Sample data
n_trials <- 100
sample_size <- 1000
sim_data_1 <- rbinom(n=sample_size, size = n_trials, prob = 0.1)
sim_data_2 <- rbinom(n=sample_size, size = n_trials, prob = 0.5)
sim_data_3 <- rbinom(n=sample_size, size = n_trials, prob = 0.8)
sim_df <- data.frame(successes = c(sim_data_1, sim_data_2, sim_data_3),
                     distribution = rep(c("p = 0.1", "p = 0.5", "p = 0.8"),
                                        each = sample_size))

# Plot the data on a histogram
ggplot(sim_df, aes(x=successes, fill = distribution)) +
  geom_histogram(binwidth = 1,
                 center = 0) +
  scale_fill_manual(values = c("p = 0.1" = "limegreen", "p = 0.5" = "tomato", "p
      = 0.8" = "royalblue")) +
  labs(title = "Comparison of Binomial Distributions",
       x = "Number of Successes",
       y = "Frequency",
       fill = "Probability (p)")
```

Binomial Distribution Plot



Applications of the Binomial Distribution

1. Quality control measures and sampling schemes.
2. Industry. In processes having two possible outcomes (Failure, Success) with constant probability of success.
3. Medicine (cure, no cure).
4. Military (the result of firing a missile (hit, miss)).

Hypergeometric Distribution

Example: Hypergeometric

Suppose we have a bag of 15 balls, of which 9 are white and 6 are black. The probability p of drawing a white ball and the probability q of drawing a black ball are:

$$p = \frac{9}{15}, \qquad q = \frac{6}{15}.$$

Hypergeometric Distribution

Example: Hypergeometric

Suppose we have a bag of 15 balls, of which 9 are white and 6 are black. The probability p of drawing a white ball and the probability q of drawing a black ball are:

$$p = \frac{9}{15}, \qquad q = \frac{6}{15}.$$

Assume we draw 5 balls from the bag. We can do this either:

(i) with replacement,

(ii) without replacement

What is the probability of drawing 3 white balls in the 5 draws?

Example: Hypergeometric

(i) With replacement :

Example: Hypergeometric

(i) With replacement :
For each draw, the probability of drawing a white ball, p, does not change. The process can be modelled by the Binomial model. Let X be the number of white balls in the 5 draws. Then

$$P(X = 3) = \binom{n}{k} p^k q^{n-k}, \qquad \text{with } n = 5,\ k = 3.$$

Example: Hypergeometric

(i) With replacement :
For each draw, the probability of drawing a white ball, p, does not change. The process can be modelled by the Binomial model. Let X be the number of white balls in the 5 draws. Then

$$P(X = 3) = \binom{n}{k} p^k q^{n-k}, \qquad \text{with } n = 5,\ k = 3.$$

(ii) Without replacement:

Example: Hypergeometric

(i) With replacement :
For each draw, the probability of drawing a white ball, p, does not change. The process can be modelled by the Binomial model. Let X be the number of white balls in the 5 draws. Then

$$P(X = 3) = \binom{n}{k} p^k q^{n-k}, \qquad \text{with } n = 5,\ k = 3.$$

(ii) Without replacement:
The probability of drawing a white ball each time changes, depending on the previous draws and the probability of drawing 3 white balls in the 5 draws follows a hypergeometric distribution.

$$P(X = 3) = \frac{\binom{9}{3}\binom{6}{2}}{\binom{15}{5}} = 0.420(3dp)$$

Hypergeometric Distribution

For a variable to follow a hypergeometric distribution, the following conditions are met.

1. A random sample of size n is selected without replacement from a population of N items (population size is N).
2. In the population k of the N items can be classified as successes, and $N - k$ items can be classified as failures (k items being of "success").
3. The size of random sample n does not exceed the number of successes k or the number of failures $N - k$, that is $n \leq \min\{k, N - k\}$ (sample size is n).

Hypergeometric Distribution

Definition

A random variable X is said to have the **hypergeometric distribution** with parameters N, n and k if

$$P(X = m) = \frac{\binom{k}{m}\binom{N-k}{n-m}}{\binom{N}{n}}, \quad m = 0, 1, 2, ..., n, \quad n \leq \min\{k, N-k\}.$$

Hypergeometric Distribution

Definition

A random variable X is said to have the **hypergeometric distribution** with parameters N, n and k if

$$P(X = m) = \frac{\binom{k}{m}\binom{N-k}{n-m}}{\binom{N}{n}}, \quad m = 0, 1, 2, ..., n, \quad n \leq \min\{k, N-k\}.$$

We write $X \sim h(N, k, n)$ and we say that X is a hypergeometric random variable.
Think of X as the **number of successes** in a random sample of size n selected from a population of size N with k successes and $N - k$ failures.

Expectation and Variance

The *expectation* and the *variance* of $X, X \sim h(N, k, n)$ are given by

$$E(X) = \frac{nk}{N}, \; Var(X) = \frac{(N-n)nk}{N(N-1)}\left(1 - \frac{k}{N}\right).$$

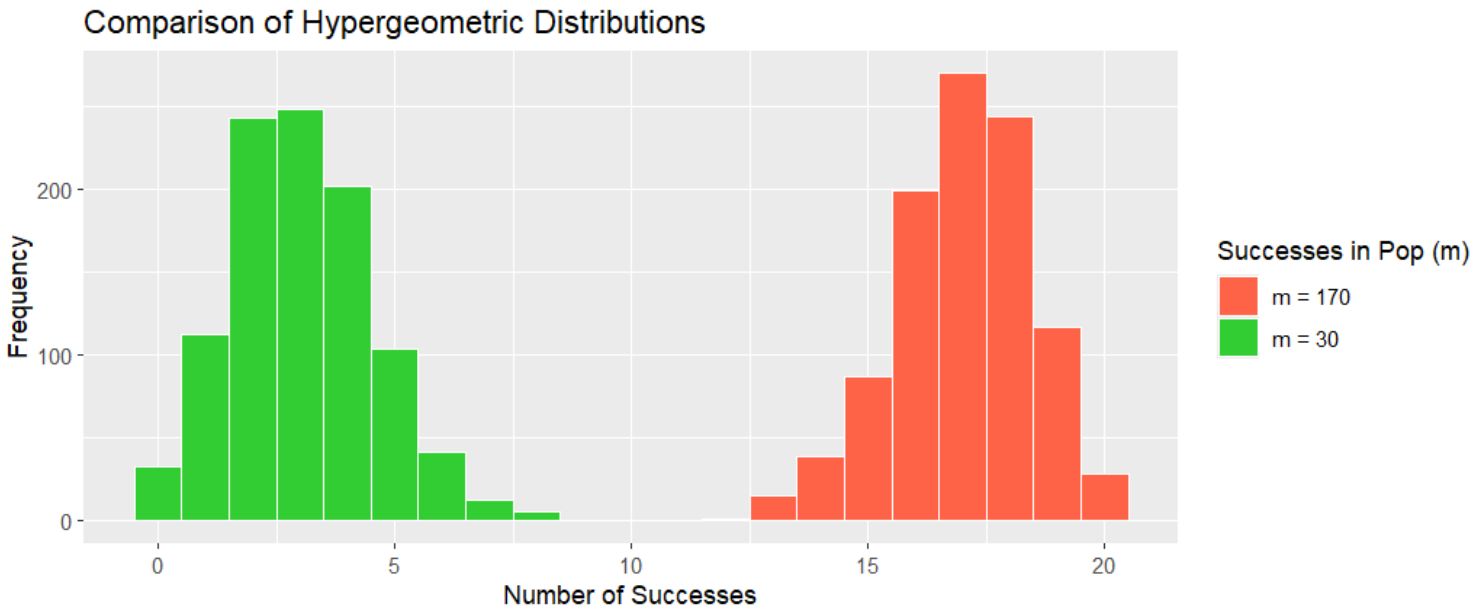
Simulation and Plotting a Hypergeometric

```
        # Sample hypergeometric data
N_pop <- 200
n_sample <- 20
sample_size <- 1000
hg_sim_data_1 <- rhyper(nn = sample_size, m = 30, n = N_pop - 30, k = n_sample)
hg_sim_data_2 <- rhyper(nn = sample_size, m = 170, n = N_pop - 170, k = n_sample
    )

sim_df <- data.frame(successes = c(hg_sim_data_1, hg_sim_data_2),
                     distribution = rep(c("m = 30", "m = 170"),
                                        each = sample_size))

\# Plot the data on a histogram
ggplot(sim_df, aes(x=successes, fill = distribution)) +
  geom_histogram(binwidth = 1,
                 center = 0,
                 color = "white") +
  scale_fill_manual(values = c("m = 30" = "limegreen", "m = 170" = "tomato")) +
  labs(title = "Comparison of Hypergeometric Distributions",
       x = "Number of Successes",
       y = "Frequency",
       fill = "Successes in Pop (m)")
```

Hypergeometric Distribution: Plot



Applications of Hypergeometric

The **applications** of hypergeometric distributions are similar to those of the binomial distribution but the probabilities of successive outcomes in a series of trials are no longer independent because a sample is drawn from a finite population without replacement.

Applications of Hypergeometric

The **applications** of hypergeometric distributions are similar to those of the binomial distribution but the probabilities of successive outcomes in a series of trials are no longer independent because a sample is drawn from a finite population without replacement.

Example: the probability of getting 3 red cards in 5 draws from a deck of 52 ordinary playing cards.

Applications of Hypergeometric

The **applications** of hypergeometric distributions are similar to those of the binomial distribution but the probabilities of successive outcomes in a series of trials are no longer independent because a sample is drawn from a finite population without replacement.

Example: the probability of getting 3 red cards in 5 draws from a deck of 52 ordinary playing cards.

Follows a binomial distribution if each card is replaced and the deck reshuffled before the next draw is made.
Follows a hypergeometric distribution if the drawn cards are not returned to the pack.

Example: Cards

Example: Cards

Find the probabilities of drawing three red cards from a deck of 52 ordinary playing cards for the two cases of (i) the cards being replaced after each draw; and (ii) the cards not being returned to the pack.

Example: Cards

Example: Cards

Find the probabilities of drawing three red cards from a deck of 52 ordinary playing cards for the two cases of (i) the cards being replaced after each draw; and (ii) the cards not being returned to the pack.

If X represents the number of red cards in 5 draws with replacement then $X \sim bin(5, 0.5)$.

$$P(X = 3) = \binom{5}{3} p^3(1-p)^2 = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^2 = \frac{5}{8} = 0.625$$

If the cards are drawn without replacement then $X \sim h(52, 26, 5)$ and the probability of selecting 3 red cards (3 red and 2 black) in 5 draws is given with

$$P(X = 3) = \frac{\binom{26}{3}\binom{26}{2}}{\binom{52}{5}} = 0.32513.$$

Acceptance Sampling

A sample of products are sampled in order to determine whether or not the entire set of products reaches an acceptable standard, e.g. on a production line.

Acceptance Sampling

A sample of products are sampled in order to determine whether or not the entire set of products reaches an acceptable standard, e.g. on a production line.

Example: Acceptance Sampling

A company groups components into sets of 40 and a set is considered to be defective if 3 or more components are defective. The set of components is tested by drawing a sample of 5 components at random. What is the probability that exactly one defective component is found in the sample, if there are 3 defective components in the entire set of 40?

Acceptance Sampling: Solution

If X represents the number of defective components in the sample of size $n = 5$, then $X \sim (40, 3, 5)$ and the asked probability is

$$P(X = 1) = \frac{\binom{3}{1}\binom{37}{4}}{\binom{40}{5}} = 0.3011.$$

Geometric Distribution

Definition

A random variable X has a **geometric** distribution with the parameter $p, 0 < p < 1$ if

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3...$$

Geometric Distribution

Definition

A random variable X has a **geometric** distribution with the parameter $p, 0 < p < 1$ if

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3...$$

We write $X \sim Geom(p)$.

Geometric Distribution

Definition

A random variable X has a **geometric** distribution with the parameter $p, 0 < p < 1$ if

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3...$$

We write $X \sim Geom(p)$.

Example: a family keeps having children until they get the first son. Assuming the probability of getting a son is 0.5, determine the probability that the family will have at least 3 children.

Expectation and Variance

Note that $\sum_{k=1}^{\infty}(1-p)^{k-1}$ is a sum of a geometric series with the common quotient 1 - p and hence

$$\sum_{k=1}^{\infty} P(X=k) = \sum_{k=1}^{\infty}(1-p)^{k-1}p = p\sum_{k=1}^{\infty}(1-p)^{k-1} = p\frac{1}{1-(1-p)} = 1$$

The expectation and variance are as follows.

$$E[X] = \frac{1}{p}, \qquad Var(X) = \frac{1-p}{p^2}.$$

Think of X as a number of trials needed to produce the first success where p is the probability of a success in each single trial.

Memoryless Property of the Geometric Distribution

The memoryless property states that if an event has not occurred by time a, the probability that it will not occur after additional time b is the same as the probability that it will not occur in time b.

Memoryless Property of the Geometric Distribution

The memoryless property states that if an event has not occurred by time a, the probability that it will not occur after additional time b is the same as the probability that it will not occur in time b.

It is independent of the fact that no events occurred during the first time period a.

$$P(X > a + b | X > a) = P(X > b)$$

The geometric distribution is the only discrete distribution with the memoryless property.

Negative Binomial Distribution

Negative binomial random variables are generalizations of geometric random variables.

Definition

A random variable X has a **negative binomial** distribution with the parameters r and $p, 0 < p < 1$, if

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, \ldots$$

Negative Binomial Distribution

Negative binomial random variables are generalizations of geometric random variables.

Definition

A random variable X has a **negative binomial** distribution with the parameters r and $p, 0 < p < 1$, if

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, \ldots$$

We write $X \sim \text{NegBin}(r, p)$ or $X \sim \text{NB}(r, p)$.

Negative Binomial Distribution

Negative binomial random variables are generalizations of geometric random variables.

Definition

A random variable X has a **negative binomial** distribution with the parameters r and $p, 0 < p < 1$, if

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, \ldots$$

We write $X \sim \text{NegBin}(r, p)$ or $X \sim \text{NB}(r, p)$.
Think of X as a number of trials needed to produce r successes where p is the probability of a success in a single trial.
The binomial coefficient $\binom{k-1}{r-1}$ gives the number of different arrangements of $(r-1)$ successes in $(k-1)$ trials.

Example: children

A couple wishes to keep having children until they have two daughters. The probability of a girl is p and the probability of a boy is $1-p$. Let X be the total number of children they have. The probability distribution for X is given by:

$$P(X=k)=(k-1)p^2(1-p)^{k-2}, \qquad k=2,3...$$

Expectation and Variance

The *expectation* and the *variance* of $X \sim \text{NegBin}(r, p)$ are

$$EX = \frac{r}{p}, \qquad VarX = \frac{r(1-p)}{p^2}.$$

Note that if $X \sim \text{NegBin}(r, p)$ then $\text{X} = \sum_i^r X_i$ where X_i are independent geometric random variables with the parameter p.

Poisson Distribution

Typically it is used to describe the number of events occurring in a fixed interval of time or space. The mathematical derivation is complicated and is hence omitted here.
Assume that an interval is divided into a very large number of equal subintervals so that the probability of the occurrence of an event in any subinterval is very small.
We assume

- The probability of the occurrence of an event is constant for all subintervals.
- There can be no more than one occurrence in each subinterval.
- Occurrences are independent; that is, an occurrence in one interval does not influence the probability of an occurrence in another interval.

Poisson Distribution

Definition

A random variable X is said to have the **Poisson** distribution with parameter λ if

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, 3, ...$$

Poisson Distribution

Definition

A random variable X is said to have the **Poisson** distribution with parameter λ if

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, 3, ...$$

We write $X \sim \text{Poisson}(\lambda)$.

Poisson Distribution: Applications, Expectation, Variance

Common applications: misprints in a book, days a university is closed due to snow during winter, arrivals into a shop, crimes in a geographical area. The expectation and variance are as follows.

$$E(X) = \lambda, \qquad Var(X) = \lambda$$

The calculations use the fact that

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}.$$

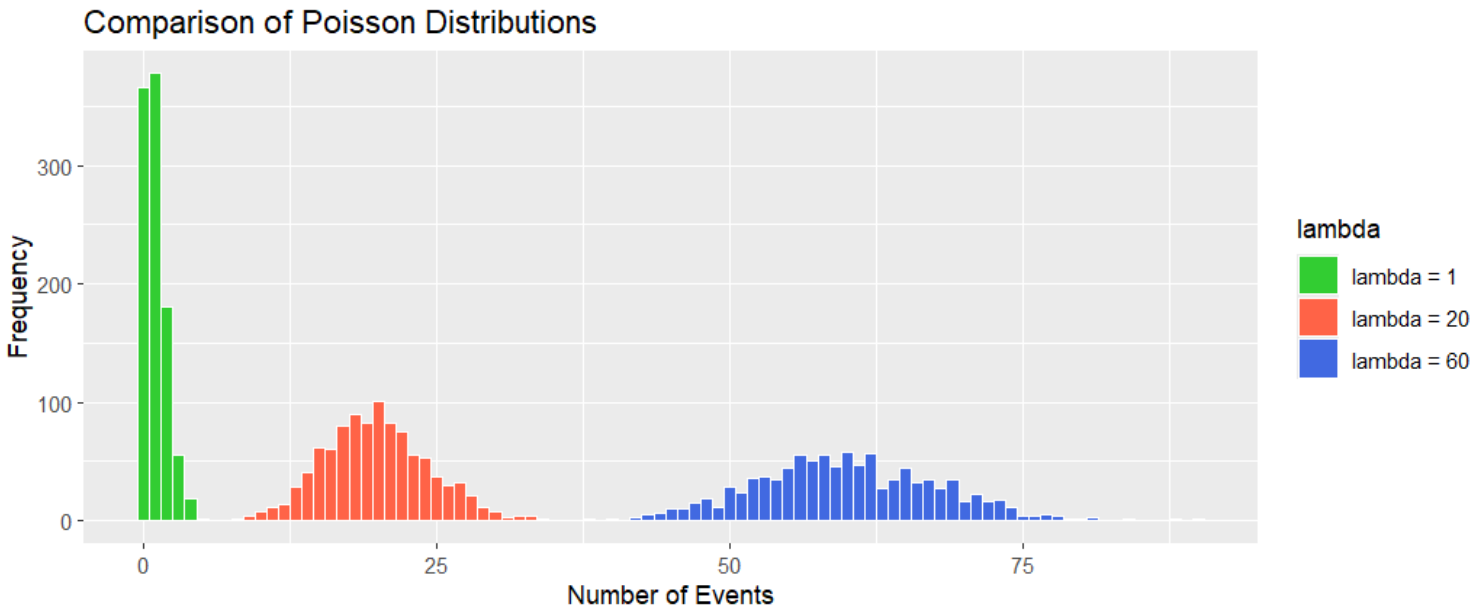
Simulating and Plotting a Poisson Distribution

```
# Sample Poisson data
sample_size <- 1000
poisson_sim_data_1 <- rpois(n = sample_size, lambda = 1)
poisson_sim_data_2 <- rpois(n = sample_size, lambda = 20)
poisson_sim_data_3 <- rpois(n = sample_size, lambda = 60)


sim_df <- data.frame(successes = c(poisson_sim_data_1, poisson_sim_data_2,
     poisson_sim_data_3),
    distribution = rep(c("lambda = 1", "lambda = 20", "lambda = 60"), each =
     sample_size))

# Plot the data on a histogram
ggplot(sim_df, aes(x=successes, fill = distribution)) +
  geom_histogram(binwidth = 1,
                 center = 0,
                 color = "white") +
  scale_fill_manual(values = c("lambda = 1" = "limegreen", "lambda = 20" = "
     tomato", "lambda = 60" = "royalblue")) +
  labs(title = "Comparison of Poisson Distributions",
       x = "Number of Events",
       y = "Frequency",
       fill = "lambda")
```

Poission Distribution: Plot



Example: Printer Error

Example: Printers

The printer on Level 4 of Building 54 has seen 5 reported failures over the last 100 days. Answer the following questions: (a) What is the probability of no failure in a given day? (b) What is the probability of one or more failures in a given day? (c) What is the probability of at least 2 failures in a 5-day period?

Example: Printer Error

Example: Printers

The printer on Level 4 of Building 54 has seen 5 reported failures over the last 100 days. Answer the following questions: (a) What is the probability of no failure in a given day? (b) What is the probability of one or more failures in a given day? (c) What is the probability of at least 2 failures in a 5-day period?

We model the number of failures (rare event) by the Poisson distribution. The expected number of failures per day is

$$\lambda = \frac{5}{100} = 0.05.$$

Printer Error Continued

(a) The probability of no failure in a given day is

$$P(\text{no failure in a given day}) = P(X = 0) = \frac{e^{-0.05}\lambda^0}{0!} = 0.9512.$$

(b) The probability of at least one failure is the complement of the probability of 0 failure:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.9512 = 0.0498.$$

(c) The average number of failures over a 5-day period is

$$\lambda = 0.05 \times 5 = 0.25.$$

Hence

$$\begin{aligned} P(X \geq 2) &= 1 - P(X \leq 1) = 1 - P(X = 0) - P(X = 1) \\ &= 1 - e^{-0.25} - \frac{e^{-0.25}0.25^1}{1!} \\ &= 1 - 0.9735 = 0.265. \end{aligned}$$

Example: Washing Machine

Example: Washing Machine

A washing machine breaks down on *average* 3 times a *year*. What is the probability of exactly 2 breakdowns during the next *year*?

Example: Washing Machine

Example: Washing Machine

A washing machine breaks down on *average* 3 times a *year*. What is the probability of exactly 2 breakdowns during the next *year*?

We know $\lambda = 3$. We can substitute this into the probability function for the Poisson distribution:

$$P(X = 2) = \frac{3^2 e^{-3}}{2} = \frac{9}{2} \times 0.04979 = 0.22405.$$

Moment Generating Function

Definition

Let X be a random variable of the discrete type with $x \in S$ being the space of all the possible values that X may take. The moment generating function (mgf) is defined as

$$M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} p(x),$$

provided that $M(t)$ is well defined (i.e., the sum does not go to infinity).

Moment Generating Functions

R1: $M(t)$ is often well defined for some range of t, and in many cases for $-\infty < t < \infty$.

R2: If $M(t)$ exists for some range of t, there is one and only one distribution of probability associated with $M(t)$. In other words, one distribution, one mgf. This is the main reason why mgfs are so important in probability theory.

Example: mgfs

If X has the mgf

$$M(t) = e^t \left(\frac{3}{8}\right) + e^{2t} \left(\frac{4}{8}\right) + e^{3t} \left(\frac{1}{8}\right), \qquad -\infty < t < \infty,$$

then the support of X is $S = \{1, 2, 3\}$ and the corresponding probabilities are

$$P(X = 1) = \frac{3}{8}, \qquad P(X = 2) = \frac{4}{8}, \qquad P(X = 3) = \frac{1}{8}.$$

Characteristics of the mgf

What makes the mgf important is the following. The first derivative is

$$M'(t) = \sum_{x \in S} x e^{tx} p(x).$$

Evaluating at $t = 0$ yields

$$M'(0) = \sum_{x \in S} x p(x) = E[X].$$

Furthermore, the second derivative is

$$M''(t) = \sum_{x \in S} x^2 e^{tx} p(x)$$

and

$$M''(0) = \sum_{x \in S} x^2 p(x) = E[X^2].$$

Characteristics of the mgf

In general,

$$M^{(r)}(0) = E[X^r].$$

This shows that we can find the moments of X by differentiating $M(t)$. In using this technique, it must be emphasized that we must first evaluate the summation representing $M(t)$ to obtain a closed form and then we differentiate that closed form to obtain the moments of X.

Example

Example: mgf

If the mgf of X is

$$M(t) = \frac{2}{5}e^t + \frac{1}{5}e^{2t} + \frac{2}{5}e^{3t},$$

find the mean and variance of X.

We differentiate $M(t)$ once to get:

$$M'(t) = \frac{2}{5}e^t + \frac{2}{5}e^{2t} + \frac{6}{5}e^{3t}$$

and

$$M'(0) = \frac{2}{5} + \frac{2}{5} + \frac{6}{5} = 2 = \mu.$$

mgf Example (Continued)

And again to get:

$$M'' = \frac{2}{5}e^t + \frac{4}{5}e^{2t} + \frac{18}{5}e^{3t}$$

and

$$M''(0) = \frac{2}{5} + \frac{4}{5} + \frac{18}{5} = \frac{24}{5}.$$

Hence,

$$\sigma^2 = E[X^2] - \mu^2 = \frac{24}{5} - 4 = \frac{4}{5}.$$

mgf for the Binomial Distribution

We calculate the mgf for the binomial distribution.

$$\begin{aligned} M(t) = E[e^{tX}] &= \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=0}^{n} \binom{n}{k} (e^t p)^k (1-p)^{n-k} \\ &= \Big(pe^t + (1-p)\Big)^n \qquad \text{(by the binomial theorem).} \end{aligned}$$

$M(t)$ exists for any $-\infty < t < \infty$.